

Mahmoud Elboghday . Jarvis Consulting

Business Intelligence Analyst with 3+ years of experience turning operational complexity into measurable business outcomes. Grounded in business fundamentals from a Hospitality Management degree and sharpened through Information Systems engineering alongside hands-on delivery at Desert Suspension, where I served as the primary data analyst across Sales, Marketing, and executive leadership at three international facilities. Identified reporting inefficiencies and built the solution end-to-end: Automated ETL workflows in Python and SQL cut manual reporting cycles by 75%, a centralized data warehouse eliminated cross-departmental silos, and statistical inventory models achieved 85% forecasting accuracy, contributing directly to a 17% net profit increase.

Skills

Proficient: Python, Machine Learning (scikit-learn, tensorflow), Data Analysis (Pandas, Numpy), SQL/RDBMS (PostgreSQL, SQL Server, Oracle), Power BI/DAX, Agile/Scrum, GCP/AWS, Git/GitHub

Competent: UML/System Design, Docker, ETL Pipeline Development, Linux/Bash, Java/Spring Boot

Familiar: Apache Spark/Databricks, Tableau, Kubernetes, React, UI/UX

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_MahmoudElboghday

Linux Cluster Resource Monitoring System [GitHub]: Designed and deployed a Linux cluster monitoring agent on Rocky Linux 9 (GCP) to give administrators real-time visibility into CPU, memory, and disk usage across distributed nodes, supporting informed decisions around resource allocation and capacity planning. Built two Bash scripts using vmstat and df to collect hardware specs and live usage metrics, persisting data into a centralized PostgreSQL instance containerized via Docker. Automated metric collection with crontab and validated end-to-end pipeline accuracy through SQL-based analysis.

Highlighted Projects

Law Enforcement BI and Resource Optimization Pipeline [GitHub]: Built an end-to-end crime analytics platform on 1.23M+ Chicago Police Department incidents (2021-2025) using a SQL Server star schema (4 dimension tables, 1 fact table) and a Python ETL pipeline maintaining 96% data completeness across critical fields. Delivered descriptive, diagnostic, predictive, and prescriptive analytics across the full stack. Trained three ML models: a Random Forest crime-type classifier (88% accuracy, top features: hour of day and severity level), a Gradient Boosting volume forecaster (R-squared 0.62, RMSE 50.66 crimes/day, MAPE 21.23%), and a DBSCAN geospatial hotspot detector (epsilon 0.01 approx. 1km radius, min samples 20) that identified 15 high-risk clusters accounting for 35% of all incidents.

Professional Experiences

Business Analyst Consultant, Jarvis (04/2026-present): Delivering BSA and data projects in an agile, sprint-based environment with structured code reviews and daily standups. Built a Linux cluster monitoring agent using Bash, PostgreSQL, and Docker to automate resource metric collection across a distributed environment.

Business Intelligence Analyst, Desert Suspension (08/2024-03/2026): Owned end-to-end portfolio analytics and operational reporting across three international facilities, serving as the primary liaison between Sales, Marketing, and executive leadership. Automated ETL workflows using Python and SQL, reducing manual reporting cycles by 75% and eliminating cross-departmental data silos. Applied statistical trend analysis to inventory data, achieving 85% forecasting accuracy and contributing to a 17% increase in net profit. Governed a centralized data warehouse maintaining integrity and usability standards across all pipelines. Produced audit-ready technical and business documentation ensuring reproducibility and knowledge continuity across reporting deliverables. Operated in an agile workflow, partnering with cross-functional stakeholders to align data outputs with strategic business priorities.

Data Analyst, Upwork (02/2023-01/2024): Delivered freelance data analytics engagements across multiple clients, producing over 10 Power BI dashboards with real-time KPI tracking that accelerated decision-making speed by 40% for 15+ stakeholders. Built automated SQL ETL pipelines consolidating multiple data sources at 98% accuracy, and designed A/B tests using Python statistical methods that drove a 15% improvement in customer engagement metrics.

Education

Sheridan College (2025-present), Advanced Diploma, Information Systems Engineering - GPA: 3.7/4.0

University of Alexandria (2015-2019), Bachelor of Science, Hotel Management and Hospitality

Miscellaneous

- IBM Data Science Professional Certificate
- DataCamp Power BI Data Analyst Certificate
- PwC Data Analytics Job Simulation
- Accenture Data Analytics Job Simulation
- Competitive athlete across multiple disciplines: started as a competitive swimmer, progressed to men's physique bodybuilding, and competed in CrossFit team events across the Middle East (Saudi Arabia and Bahrain).
- Currently focused on staying active and healthy as a new father, training to keep up with my daughter as she grows.
- Content creator with over 300k followers covering Canadian immigration topics, licensing process for foreign-trained (physicians, dentists, pharmacists, and engineers) by interviewing Licensed fellows who went through the journey who completed the process and share their experience to guide others.